

A Mathematical Framework of Housing

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Abstract

Housing is simultaneously a physical asset, a financial asset, collateral, consumption capital, and a legal institution. This paper proposes a compact mathematical framework integrating housing lifecycle dynamics, mortgage finance, ownership, legal status, and occupancy into a unified state-space representation. Instead of deriving every component from first principles, the framework synthesizes ideas from housing economics, finance, macroeconomics, stock-flow consistent modelling, Markov processes, network theory, control theory, and computational economics.

The paper ends with “The End”

1 Introduction

Housing occupies a unique position in modern economies. It represents one of the largest household assets, a major source of bank lending, an important determinant of macroeconomic fluctuations, and a central component of welfare.

Previous research has examined housing from multiple perspectives, including urban economics, household finance, mortgage markets, asset pricing, stock-flow consistent macroeconomics, dynamic programming, and financial stability. This paper proposes a compact mathematical framework capable of encompassing these perspectives within a common state-space representation.

2 Housing States

Every housing unit occupies one of five observable lifecycle states:

State	Description
1	Not yet constructed
2	Constructed, mortgage-free
3	Constructed with mortgage
4	Constructed with reverse mortgage
5	Foreclosed after default

Let

$$S_i(t) \in \{1, 2, 3, 4, 5\}$$

denote the state of house i .

Define

$$N_j(t) = \#\{i : S_i(t) = j\}.$$

The total housing stock satisfies

$$N = \sum_{j=1}^5 N_j.$$

3 A State-Space Representation

Rather than treating the five states as primitive, define the complete housing state as

$$X_i = (S_i, F_i, O_i, L_i, Q_i),$$

where

S_i structural state,

F_i financial state,

O_i ownership state,

L_i legal state,

Q_i occupancy state.

The original five categories are special cases obtained by particular combinations of these variables.

4 Financial State

Let

$$F_i = (M_i, R_i, E_i),$$

where

M_i mortgage balance,

R_i reverse mortgage balance,

E_i homeowner equity.

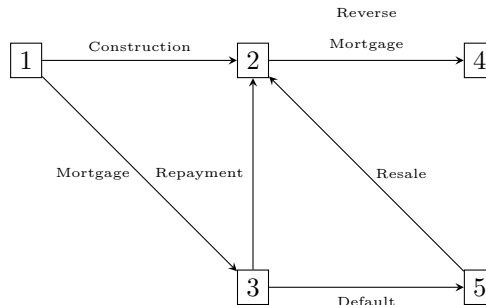
The accounting identity

$$P_i = M_i + R_i + E_i$$

decomposes housing value into debt and equity.

5 Housing Dynamics

Housing evolves through discrete transitions.



These transitions may be represented by a Markov transition matrix or continuous-time generator.

6 Conservation Principles

The physical housing stock obeys

$$\frac{dN}{dt} = C - D,$$

where C denotes construction and D demolition.

Housing capital satisfies

$$\frac{dK}{dt} = I - \delta K.$$

Mortgage accounting obeys

$$\sum_i M_i = \sum_j M_j^{Bank},$$

while reverse mortgages satisfy

$$\sum_i R_i = \sum_j R_j^{Bank}.$$

These identities express stock-flow consistency.

7 General Equilibrium

The housing market consists of interacting markets:

Market	Principal Object
Land	Land parcels
Construction	New housing
Housing	Existing homes
Mortgage	Mortgage contracts
Reverse Mortgage	Equity release
Rental	Occupancy rights
Foreclosure	Distressed housing
Secondary Finance	Mortgage-backed claims

A stationary equilibrium simultaneously satisfies

- housing market clearing,
- credit market clearing,
- household optimisation,
- developer optimisation,
- bank solvency,
- government policy constraints.

8 A Computational View

The state-space representation naturally supports

- agent-based simulation,
- dynamic programming,
- reinforcement learning,

- graph algorithms,
- network-flow optimisation,
- Monte Carlo simulation,
- Bayesian estimation,
- hidden Markov models.

The modular structure also permits extensions involving climate risk, energy efficiency, zoning regulations, insurance, taxation, and demographic transitions.

Proposition 1. *The multidimensional state-space strictly generalises the original five-state housing model.*

Proof. The original classification specifies only a finite set of observable lifecycle categories. Every such category corresponds to a particular combination of structural, financial, ownership, legal, and occupancy variables. Conversely, the multidimensional state-space permits additional distinctions without altering the underlying mathematical architecture. Hence the original model is embedded as a special case. $\square$

9 Interdisciplinary Perspective

The proposed framework draws upon several disciplines.

Field	Contribution
Economics	Housing demand and supply
Finance	Mortgages and collateral
Mathematics	State-space modelling
Statistics	Estimation and forecasting
Computer Science	Algorithms and simulation
AI/ML	Reinforcement learning and prediction
Political Science	Housing policy
International Relations	Cross-country housing systems
Psychology	Household decision-making
Public Health	Housing quality and wellbeing
Urban Planning	Land use and zoning

10 Conclusion

Viewing housing as a multidimensional state-space unifies physical housing capital, financial contracts, legal institutions, ownership structures, and occupancy decisions. The framework accommodates lifecycle transitions, stock-flow consistency, general equilibrium analysis, and computational implementation while remaining flexible enough to incorporate future institutional and technological developments.

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Glossary

S_i	Housing lifecycle state.
F_i	Financial state.
O_i	Ownership state.
L_i	Legal state.
Q_i	Occupancy state.
M_i	Mortgage balance.
R_i	Reverse mortgage balance.
E_i	Homeowner equity.
K	Housing capital.
N	Housing stock.

The End